

Diesel engine

16V4000M60R

Spare parts catalog for product no.

527101451

Printed in Germany

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Subject to alterations and amendments.

Preface

This catalog is published for parts requisitioning purposes and provides assistance in engine disassembly and reassembly. It contains a complete list of all engine components, including those which are not available as replacement parts.

The parts originate from the Engine Design File and document the product's construction status.

Catalog Arrangement

The catalog is subdivided into functional groups and functional group segments (FG/FGS).

The headings of the illustration pages contain the following information:

- ◆ TYPE
- ◆ SERIAL NO.
- ◆ FG/FGS DESTINATION
- ◆ FG/FGS NO.
- ◆ TYP

Illustrations

Explanation of Columns

- ◆ POS: Provides a cross-reference from illustration to text section. Non-illustrated parts are indicated by the suffix "Z" after the C/N.
- ◆ PART NO.: This is the valid number for ordering the associated part.
- ◆ DESIGNATION: The precise part name.
- ◆ QTY. (Quantity of Unit): This column indicates the quantity of this part required for the functional group.
- ◆ MEASUREMENT
- ◆ FN (Footnote): Provides additional parts information. For clarification of footnote symbols, see Page 3/2.
- ◆ REMARKS: This column is intended for supplementary information.

Amendments Service

Part No. changes and amendments which occur after publication of the catalog will not be published. We therefore recommend updating of the catalog part numbers by the end user. This information is provided, for instance, in order confirmations and/or on equipment labels.

Footnote Symbols Identification

- A Index footnote
- AA See separate manual
- AB Supplied as a complete assembly. Repair by manufacturer only.
- AC No part number assigned. State illustration table number, callout number, designation and catalog number when ordering.
- AD Quantity referenced to one unit.
- E Magnetically treated part or assembly (stamped
- EM Emission-relevant parts. In the case of all marine engines with IMO certification it should be noted that identification number of the emission-relevant parts on the engine must agree with the details given in the technical file. Give the EM code also when ordering with the order No., e.g. 535 030 04 17 EM.
- K Following reasons require ordering of complete parts group:
- KA Part machined jointly with other components
- KB Inseparable connection
- KC Machining on group assembly
- KD Group balancing required
- KE Individual parts not obtainable through the parts supply system
- KF Replacement part will be assigned a different part number
- KG Available separately

- M** Parts having different repair sizes.
Dimensions are listed in the Tolerances and Wear Limits List.
- NB** As required
- P** Bought-out components
- S** Welded - Parts welded together to form an assembly
Component parts are not listed as they are not included in the parts supply system
- SA** Component part of a welded assembly obtainable through the parts-supply system. Replacement requires welding or brazing
- SB** Group of parts which are welded or bolted together.
- U** Part whose dimensions are determined on assembly.
- UA** Indicate dimensions when ordering
- UB** Supplied as a complete set only
- UC** Part requires shaping/bending on assembly
- WW** Optional
- X**
- Y** Symbols used for footnotes appearing on the catalog page.
- Z**

Ordering Procedure

The following information is essential for spare-part order processing:

- ◆ **TYPE**
- ◆ **SERIAL NO.**
- ◆ **FG/FGS**
- ◆ **BOM**
- ◆ **Pos. number**
- ◆ **Designation**
- ◆ **Quantity**
- ◆ **Page**

When ordering replacement parts for ancillary equipment such as injection pumps

Phone orders should be confirmed in writing to prevent errors.

Ordering Example:

Factory Repairs

Parts shipped to MTU for repair should have an engine serial-number identification (tag or label, for example).

Exchange Components (Reman)

Exchange components are parts overhauled by the manufacturer

New component in exchange procedure

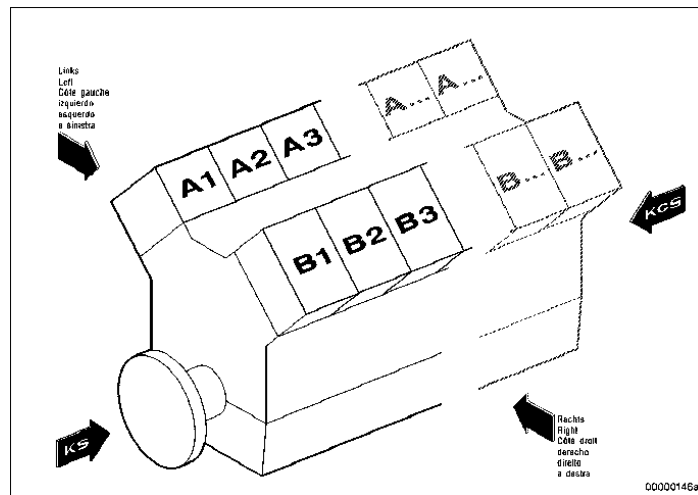
Reman component in exchange procedure

Exchange Procedure

With the introduction of exchange components

For further information refer to Service Information 01/S0028.

Engine Side And Cylinder Designations



Engine sides are always designated as viewed from driving end.

The left bank of cylinders is marked

The cylinders of each bank are numbered consecutively

Numbering of other engine components is also from the drive end

Designations and Abbreviations

Driving end = KS

Free end = KGS

Left side

Right side

Upper side

Lower side

Turbocharger = ATL

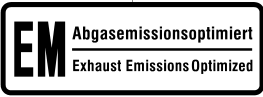
Overview functional groups

FG	FG DESIGNATION
004	LABELS AND MISCELLANEOUS MATERIALS
011	CRANKCASE
013	CYLINDER LINER
014	OIL PAN
015	ENGINE MOUNTING
016	END HOUSING, MAIN PTO END AND AUX PTO END
018	CRANKCASE BREATHER
024	GEARCASE, AUX PTO END
025	GEAR TRAIN, AUX PTO END
031	CRANKSHAFT ...
032	CRANKSHAFT SUPPORT BEARINGS, MAIN PTO END
035	CRANKSHAFT, PTO, AUX PTO END
036	CRANKSHAFT TRANSPORTATION LOCKING DEVICE
037	PISTON
038	CONROD
041	CYLINDER HEAD
042	CYLINDER HEAD ATTACHMENTS
051	CAMSHAFT
052	CAMSHAFT DRIVE
053	CAMSHAFT AXIAL ALIGNMENT BEARING
054	ROLLER TAPPETS, PUSH RODS
055	VALVE DRIVE
073	INJECTION PUMP / HP PUMP
075	FUEL INJECTOR
077	HP FUEL LINE / HP LINE
081	FUEL PIPEWORK BEFORE DELIVERY PUMP
082	PIPEWORK BETWEEN DELIVERY PUMP AND FILTER
083	FUEL FILTER
084	FUEL SUPPLY TO INJECTION PUMP
086	LEAK-OFF FUEL LINE
087	LEAK-OFF FUEL MONITORING
101	TURBOCHARGER (H.P)
111	CHARGE AIR COOLING, IF REQUIRED, LEFT SIDE
113	CHARGE AIR COOLING, RIGHT SIDE
121	AIF FILTER
123	AIR SYSTEM
124	AIR SUPPLY TO CYLINDERS
126	AIR TRANSFER, LEFT - RIGHT SIDE, MAIN PTO END
128	CONTROL BLOCK FOR AIR AND EXHAUST FLOW CONTROL FLAPS
142	CONSTANT-PRESSURE EXHAUST MANIFOLD
143	CARRIER HOUSING
144	EXHAUST FLOW CONTROL FLAP WITH CONTROLS
146	EXHAUST SYSTEM AFTER ENGINE
172	ELECTRIC STARTING
181	LUBE OIL PUMP AND DRIVE
183	OIL FILTRATION/COOLING
184	CRANKCASE-INTERNAL OIL SUPPLY

Overview functional groups

FG FG DESIGNATION

185	TURBOCHARGER OIL SUPPLY
188	CONNECTIONS FOR ENGINE OIL SUPPLY
189	OIL REPLENISHMENT PUMP
202	HIGH-TEMPERATURE CIRCUIT
203	LOW-TEMPERATURE CIRCUIT
205	COOLANT COOLER ATTACHING H/WARE
206	COOLANT PIPEWORK WITH THERMOSTAT
207	RAW WATER PUMPE WITH CONNECTIONS
208	COOLANT EXPANSION TANK
213	GENERATOR
231	ENGINE MOUNTING
251	STEEL-SPRING COUPLING
255	BENDING-RESILIENT COUPLING
342	DRIVE SYSTEM
500	MONITORING AND CONTROL SYSTEM, GENERAL ELECTRICAL EQUIPMENT
507	GENERAL WIRING, ENGINE/GEARBOX/PLANT
508	ENGINE GOVERNOR (ELECTRONIC) CONTROL SYSTEM
511	ALARM SYSTEM
519	LEVEL MONITORING SYSTEM
526	REMOTE CONTROL
535	TEST AND MAINTENANCE EQUIPMENT



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INSTALLATION OF LABELS

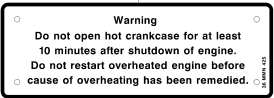
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MISCELLANEOUS MATERIALS (MTN5012)

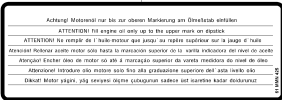
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MISCELLANEOUS MATERIALS (MTN5012)

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MISCELLANEOUS MATERIALS (MTN5012)

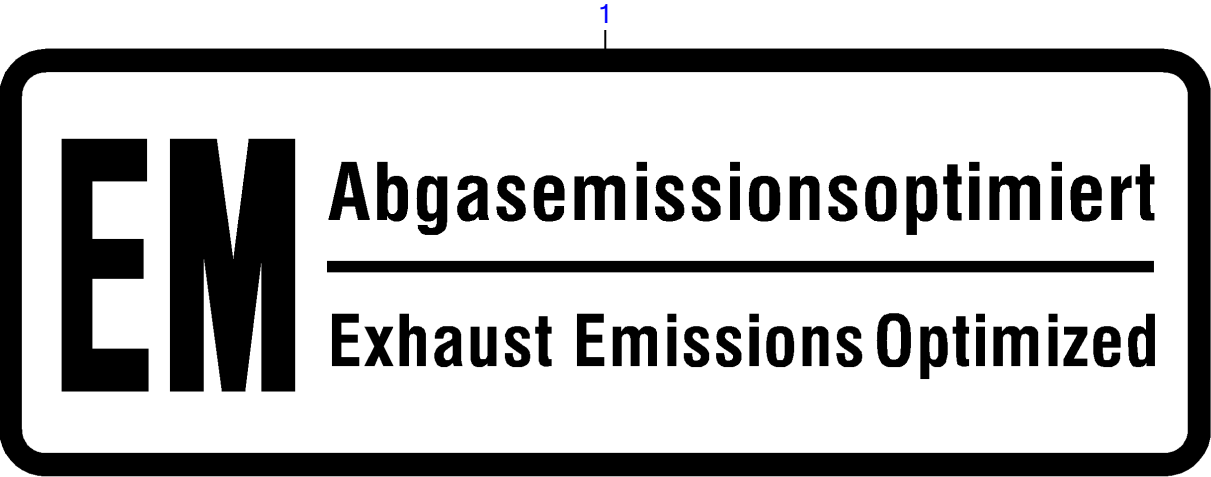
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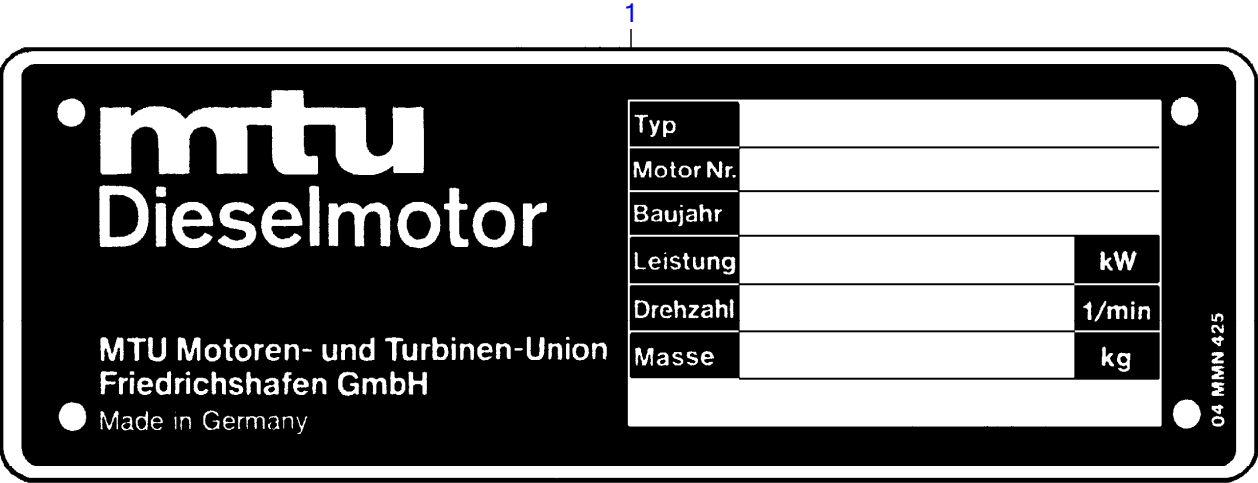
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POS.	PART NO.	DESIGNATION	QTY.	MEASUREMENT
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